



Approximation algorithm for a generalized Roman domination problem in unit ball graphs

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Abstract

In this paper we propose a generalized Roman domination problem called connected strong k -Roman dominating set problem. It is NP-hard even in a unit ball graph. Unit ball graphs are the intersection graphs of equal sized balls in the three-dimensional space, they are widely used as a mathematical model for wireless sensor networks and some problems in computational geometry. This paper presents the first constant approximation algorithm with a guaranteed performance ratio at most $6(k+2)$ in unit ball graphs, where k is a positive integer.

Keywords Connected strong k -Roman dominating set · Constant approximation algorithm · Unit ball graph

1 Introduction

In articles in *Scientific American* by Stewart entitled “Defend the Roman Empire!” (Stewart 1999) and in *Discrete Mathematics* by Cockayne et al. entitled “Roman domination in graphs” (Cockayne et al. 2004), the concept of Roman domination was introduced. Each vertex in the graph model corresponds to a location in the Roman Empire. A location (vertex v) is considered unprotected if no legions are stationed

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there (i.e., $f(v) = 0$) and protected otherwise (i.e., if $f(v) \in \{1, 2\}$). A location (vertex v) having no legion can be protected by a legion sent from a neighboring location (an adjacent vertex u). Moreover, Emperor Constantine the Great (Cockayne et al. 2004) decreed that a legion cannot be sent to a neighboring location if it leaves its original station unprotected. In other words, every location with no legion must be adjacent to a location that has at least two legions. The formal definition of this defense strategy was presented in Cockayne et al. (2004). A *Roman dominating function* on a graph $G = (V, E)$ is defined as a function $f : V \mapsto \{0, 1, 2\}$ satisfying the condition that every vertex u with $f(u) = 0$ has at least one neighbor v with $f(v) = 2$. The weight of f is $\sum_{u \in V} f(u)$. The *Roman domination number* $\gamma_R(G)$ is the minimum weight of a Roman dominating function of G . Since it is expensive to maintain a legion at a location, the Emperor would like to station as few legions as possible, while still defending the Roman Empire. To defend the Roman Empire from single attack, many graph theoretic models have been studied, for example, in Cockayne et al. (2004), Chambers et al. (2009), Henning and Hedetniemi (2003) and Liu and Chang (2012). If the Roman Empire was attacked by more than one enemy army, how should Emperor Constantine the Great station the legions to defend the Roman Empire? Henning (2003) proposed a generalization of Roman dominating function to deal with multiple attacks. In this paper we propose a similar but different generalization and present an approximation algorithm.

Our mathematical model is a connected graph $G = (V, E)$ with vertex set V and edge set E . Each vertex in our graph represents a location in the Roman Empire, and two vertices are joined by an edge if the corresponding locations are adjacent. Let k be a positive integer and f be a function $f : V \rightarrow \{0, 1, \dots, k+1\}$. A location (vertex v) is considered *unprotected* if no legions are stationed there (i.e., $f(v) = 0$) or *protected* (i.e., if $f(v) \in \{1, 2, \dots, k+1\}$), otherwise. For $i = 0, 1, \dots, k+1$, let V_i be the set of vertices assigned value i under f . Note that there is a 1-1 correspondence between the functions $f : V \rightarrow \{0, 1, \dots, k+1\}$ and the ordered partitions $(V_0, V_1, \dots, V_{k+1})$ of V . Thus we will write $f = (V_0, V_1, \dots, V_{k+1})$.

We say that a vertex in V_0 is undefended with respect to f , if it is not adjacent to a vertex in V_i for any $i \geq 2$. We say that f has no undefended vertex if no vertex in V_0 is undefended with respect to f . If a vertex $u \in V_0$ is adjacent to a node $v \in V_i$ for some $i, 2 \leq i \leq k+1$, then we say that the function $g : V \rightarrow \{0, 1, \dots, k+1\}$, defined by $g(u) = 1, g(v) = f(v) - 1$ and $g(w) = f(w)$ if $w \in V - \{u, v\}$, is obtained from f by one *movement* from v to u .

Definition 1.1 (*strong k -Roman dominating function*) Let $f : V \mapsto \{0, 1, \dots, k+1\}$ be a “deployment function”. We call the function f a *strong k -Roman dominating function* if f has no undefended vertex, and for arbitrary sequence of distinct vertices $v_1, v_2, \dots, v_k$, there exists a sequence of functions $f = f_0, f_1, \dots, f_k$ such that for $i = 1, \dots, k$,

- (i) either $f_{i-1}(v_i) > 0$, in which case $f_i = f_{i-1}$,
- (ii) or $f_{i-1}(v_i) = 0$, in which case v_i has a neighbor u with $f_{i-1}(u) \geq 2$ such that $f_i(u) = f_{i-1}(u) - 1, f_i(v_i) = f_{i-1}(v_i) + 1$, and $f_i(w) = f_{i-1}(w)$ for $w \in V - \{u, v_i\}$. In other words, f_i is obtained from f_{i-1} by one movement to v_i , the movement comes from a node u with $f_{i-1}(u) \geq 2$,

(iii) f_i has no undefended vertex.

If f_i is obtained from f_{i-1} by one movement for each $i = 1, \dots, k$, then we say that f_k is obtained from f_0 by k movements.

The above definition is different from the k -Roman dominating function given by Henning in Henning (2003). Henning requires that every vertex u with $f_i(u) = 0$ has at least one neighbor v with $f_i(v) \geq 1$ for any function f_i , $i = 1, 2, \dots, k+1$. When $k = 1$, the k -Roman dominating function defined by Henning (2003) is the weak Roman dominating function studied in Henning and Hedetniemi (2003) and Zhu and Shao (2018). For convenience of the readers, we present the definition of the weak Roman dominating function as follows. Let $G = (V, E)$ be a graph and f be a function $f : V \mapsto \{0, 1, 2\}$. A vertex u with $f(u) = 0$ is said to be undefended with respect to f if it is not adjacent to a vertex with positive weight. The function f is a weak Roman dominating function (WRDF) if each vertex u with $f(u) = 0$ is adjacent to a vertex v with $f(v) > 0$ such that the function $f' : V \mapsto \{0, 1, 2\}$, defined by $f'(u) = 1$, $f'(v) = f(v) - 1$ and $f'(w) = f(w)$ if $w \in V - \{u, v\}$, has no undefended vertex. However, our definition requires that each vertex u with $f_i(u) = 0$ has at least one neighbor v with $f_i(v) \geq 2$ for any function f_i , $i = 1, 2, \dots, k-1$. This is why we call it a strong k -Roman dominating function in order to distinguish it from the one (Henning 2003). When $k = 1$, the strong k -Roman dominating function given by Definition 1.1 is the classic Roman dominating function.

Definition 1.2 ((connected) strong k -Roman dominating set) For a strong k Roman dominating function f of graph G , denote $V_i^f = \{v \in V : f(v) = i\}$ for $i = 0, 1, \dots, k+1$. Call $D_R^k(f) = (V_1^f, V_2^f, \dots, V_{k+1}^f)$ the strong k -Roman dominating set associated with f , whose weight $w(D_R^k(f)) = \sum_{u \in V(G)} f(u) = \sum_{i=1}^{k+1} i|V_i^f|$. This weight is also called the weight of f and denoted as $w(f)$. The strong k -Roman domination number $\gamma_R^k(G)$ is the weight of a minimum strong k -Roman dominating set of graph G . A strong k -Roman dominating set with weight $\gamma_R^k(G)$ is called a $\gamma_R^k(G)$ -set.

If the subgraph of G induced by $D_R^k(f)$ is connected, then f is a connected strong k -Roman dominating function and $D_R^k(f)$ is a connected strong k -Roman dominating set (which is denoted as $D_{CR}^k(f)$). The concepts of connected strong k -Roman domination number $\gamma_{CR}^k(G)$ and $\gamma_{CR}^k(G)$ -set can be defined similarly as above.

The minimum (connected) strong k -Roman dominating set problem is to compute a $(\gamma_{CR}^k(G)$ -set) $\gamma_R^k(G)$ -set for given graph G .

Notice that the minimum (connected) strong k -Roman dominating set problem is equivalent to computing a minimum weight (connected) strong k -Roman dominating function f . We shall not distinguish them in the following.

The definition of the strong k -Roman dominating function is motivated as follows. Since an undefended location is vulnerable to an attack, we require that every protected location cannot send a legion to a neighboring location if it leaves its original station unprotected. Put it in another way, every unprotected location can be defended without creating an undefended location. We make the assumption that the Roman Empire is attacked by more than one enemy army, if any, multiple attacks are consecutive

(and do not occur simultaneously). In this case, how can Emperor Constantine the Great save substantial costs of maintaining legions while still defending the Roman Empire from k consecutive attacks? Hence, how to find a method of placing the legions corresponds to how to find a strong k -Roman dominating function, and computing a most effective placement of legions corresponds to computing a minimum strong k -Roman dominating function.

Despite of its origination in guarding the Roman empire, the Roman dominating set problem also has many applications in wireless sensor networks, for example, how can one collect data timely by placing as few sensors as possible in the network (Shang and Hu 2007). Unit disk graphs are widely adopted models for homogeneous wireless networks. Shang and Hu (2007) presented an approximation algorithm for Roman dominating set in unit disk graphs. But in many real applications, a unit disk graph is not sufficient. There are cases in which three-dimensional models are needed, such as under-water sensor systems, outerspace sensor systems, etc (Akyildiz et al. 2005). In such settings, deployed sensors would form a unit ball graph. In this paper, we study connected strong k -Roman dominating set in unit ball graphs.

1.1 Related work

If $k = 1$, then the problem is exactly the minimum (connected) Roman dominating set problem, for which there are a lot of studies.

It is known that the minimum Roman dominating set problem is NP-hard (Dreyer 2000). In Pagourtzis et al. (2002), a $(2+2 \ln n)$ -approximation algorithm was proposed for general graphs, and a polynomial time approximation scheme (PTAS) was proposed for planar graphs. Lideloff et al. (2005) showed that the Roman domination number of an interval graph can be computed in linear time. Shang and Hu (2007) presented a simple linear time approximation algorithm and a polynomial-time approximation scheme for the Roman domination problem in unit disk graphs.

The definition of k -Roman dominating set is given by Henning (2003). In Henning (2003), Henning showed that for any graph G and any integer $k \geq 1$, $\gamma(G) \leq \gamma_r^k(G) \leq (k+1)\gamma(G)$, investigated graphs G with $\gamma(G) = \gamma_r^k(G)$ and characterized trees T for which $\gamma(T) = \gamma_r^k(T)$, where $\gamma(G)$ and $\gamma_r^k(G)$ denote the minimum domination number and the minimum k -Roman domination number of G , respectively. Furthermore, he investigated graphs G with $\gamma(G) = (k+1)\gamma_r^k(G)$ and characterized trees T for which $\gamma(T) = (k+1)\gamma_r^k(T)$.

Connectivity constraints come in naturally, e.g., in many applications concerning wireless sensor networks, in which it is usually important to ensure connectivity if the sensor devices have limited capabilities of computation, energy and communication, and still want to share information. Shang et al. (2010) proved that both the minimum Roman dominating set problem and the minimum connected Roman dominating set problem are NP-hard even in unit disk graphs. In addition, they presented a 5-approximation algorithm for minimum Roman dominating set problem and a 7.5-approximation algorithm for minimum connected Roman dominating set problem in unit disk graphs.

Based on the above definitions and descriptions, when $k = 1$, the k -Roman dominating function given by Henning (2003) is the weak Roman dominating function, but our function is the classic Roman dominating function. On the other hand, Henning (2003) mainly considered the k -Roman dominating set problem from the graph theoretical point of view. While we study the strong k -Roman dominating set problem from the perspective of the approximation algorithms in this paper, and give a constant approximation algorithm for the minimum connected strong k -Roman dominating set problem in unit ball graphs.

Properties of Roman dominating functions are also studied from the graph theoretical point of view. Interested readers may refer to Cockayne et al. (2004), Chambers et al. (2009), Lidelloff et al. (2005) and Liu and Chang (2012).

1.2 Our contribution

Our major contributions are summarized as follows:

1. We propose the connected strong k -Roman dominating set for the purpose of modeling the defence problem of Roman Empire in the presence multiple attacks.
2. We present a $6(k + 2)$ -approximation algorithm for the minimum connected strong k -Roman dominating set problem in unit ball graphs. Since the minimum connected Roman dominating set problem in a unit disk graph is NP-hard (Shang et al. 2010) and because a unit disk graph is a special unit ball graph, our problem is also NP-hard. Our algorithm is the first approximation algorithm on this problem achieving constant performance ratio.

The paper is organized as follows. In Sect. 2, we design an approximation algorithm for the minimum connected strong k -Roman dominating set problem in unit ball graphs, prove its correctness, analyze its time complexity, and derive its performance ratio. Section 3 concludes the paper.

2 A constant approximation algorithm

Given a graph $G = (V, E)$, let $N(u) = \{v : uv \in E\}$ and $N[u] = N(u) \cup \{u\}$ be the *open neighborhood* and the *closed neighborhood* of u in G , respectively. The degree of vertex u in G is $d(u) = |N(u)|$. For a vertex subset $S \subseteq V$, let $N(S) = (\bigcup_{u \in S} N(u)) \setminus S$ be the *neighbor set* of S , $N[S] = N(S) \cup S$ be the *closed neighbor set* of S , and denote by $G[S]$ the subgraph of G induced by S .

Definition 2.1 (*Unit ball graph*) Let $V \subset \mathbb{R}^3$ be a set of vertices in the three-dimensional space. Graph $G = (V, E)$ is a *unit ball graph* if two vertices are adjacent in G if and only if their Euclidean distance is at most one.

Definition 2.2 (*Independent set*) A subset $I \subseteq V$ is an *independent set* in graph G if no vertices in I are adjacent in G . Moreover, I is a *maximal independent set* (MIS) if $I \cup \{u\}$ is no longer an independent set for any vertex $u \in V \setminus I$. A *maximum independent set* (MaxIS) is an MIS of the largest size.

Proposition 2.3 (Huang et al. 2004) *In a unit ball graph $G = (V, E)$, for any vertex $u \in V$, there are at most 12 independent vertices in $N[u]$.*

2.1 Description of the algorithm

The algorithm is formally described in Algorithm 1. Initially, all vertices are unlabelled. Then, labels are given to vertices step by step. The label $g(u)$ of vertex u indicates how many mobile sensors are placed at u . For a vertex u , $C(u)$ is used to record the set of *children* of u . Every time a vertex u is given a label 0, it is assigned a *parent* $P(u)$, which will be used in designing the strategy of movements. During the execution of the algorithm, $un(u)$ denotes the number of currently unlabeled vertices in $N[u]$. It will be shown in next subsection that, at the end of the first while loop, all those vertices with positive labels will form a strong k -Roman dominating set. Then, more vertices are added to connect them. This step is accomplished by changing the labels of some vertices from 0 to 1.

Algorithm 1 Algorithm for the connected strong k -Roman dominating set problem.

Input: A connected unit ball graph $G = (V, E)$ and a positive integer $k \geq 1$.

Output: A connected strong k -Roman dominating set D_{CR}^k .

```

1: Choose a vertex  $u_1 \in V$  with the maximum degree, set  $g(u_1) \leftarrow \min\{d(u_1) + 1, k + 1\}$  and  $C(u_1) \leftarrow N(u_1)$ .
2: For every vertex  $u \in N(u_1)$ , set  $g(u) \leftarrow 0$ ,  $P(u) \leftarrow u_1$ .
3: while  $\exists$  an unlabeled vertex  $u$  which is adjacent to a vertex with label 0 do
4:   Set  $g(u) \leftarrow \min\{un(u), k + 1\}$ , and let  $C(u)$  be the set of unlabeled vertices in  $N(u)$ .
5:   if  $g(u) \geq 2$  then
6:     For every vertex  $v \in C(u)$ , set  $g(v) \leftarrow 0$ ,  $P(v) \leftarrow u$ .
7:   end if
8: end while
9: Let  $D_R^k$  be the set of vertices with positive labels. Set  $D \leftarrow D_R^k$ .
10: while  $G[D]$  is not connected do
11:   Let  $R$  be the connected component of  $G[D]$  containing  $u_1$ . Choose a vertex  $u$  with label 0 which is adjacent to  $R$  and another connected component of  $G[D]$ . Change the label of  $u$  to 1.
12:   Set  $D \leftarrow D \cup \{u\}$ .
13: end while
14: return  $D_{CR}^k \leftarrow D$ .
```

Figure 1 illustrates the execution of the algorithm, in which $k = 4$. (a) is the topology of the network. In (b), vertex u_1 is selected because it has the maximum degree, and $f(u_1) = \min\{d(u_1) + 1, k + 1\} = 5$, $C(u_1) = \{u_2, u_3, u_5, u_6, u_7\}$. All vertices in $C(u_1)$ are labeled by 0. In (c), vertex u_{10} is selected because it is an unlabeled vertex which is adjacent to vertex u_7 with label 0. Since $un(u_{10}) = 1$, we have $f(u_{10}) = 1$ and $C(u_{10}) = \emptyset$. In (d), vertex u_8 is selected. Since $un(u_8) = 3$, we have $f(u_8) = \min\{un(u_8), k + 1\} = 3$ and $C(u_8) = \{u_9, u_{11}\}$. Proceeding in this way, the labels after the first phase is shown in (e). In the second phase, labels of vertices u_3, u_7, u_9, u_{13} are changed from 0 to 1 sequentially. The final connected strong 4-Roman dominating set is shown in (f).

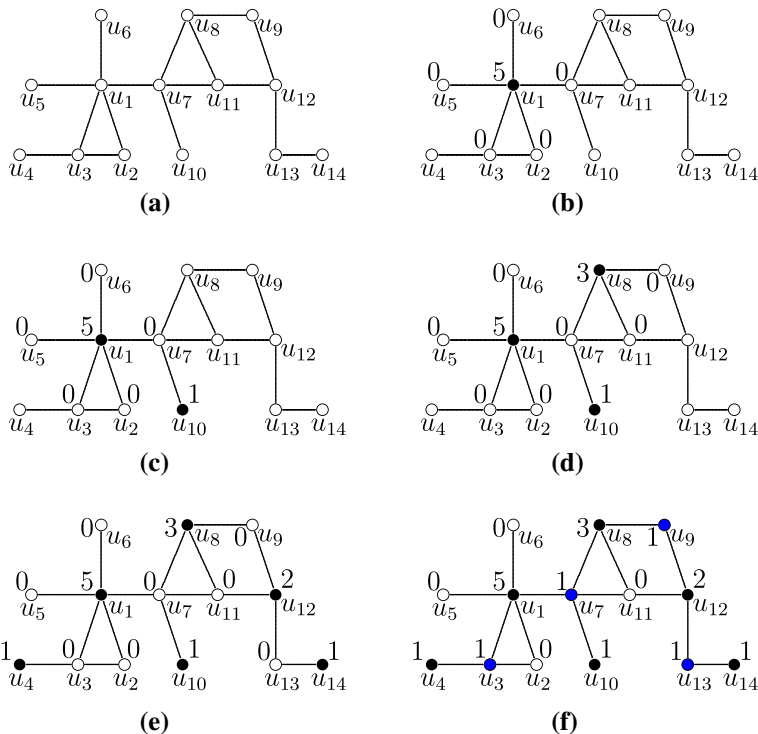


Fig. 1 An illustration of the algorithm, in which $k = 4$

2.2 Analysis of the algorithm

In this subsection, we show the correctness of Algorithm 1 and analyze its running time as well as its performance ratio. First, some auxiliary lemmas are given before addressing the main theorem.

Lemma 2.4 *When the algorithm jumps out of the first while loop, all vertices are labeled.*

Proof It suffices to show that if there exists an unlabeled vertex, then the unlabeled vertex must be adjacent to a vertex with label 0, and thus the algorithm will not jump out of the first while loop. In fact, if there exists an unlabeled vertex v , then by the connectedness of the graph, the vertex v must be adjacent to some labeled vertices. Assume that u is the first neighbor of vertex v which has received its label. If u has label 0, then we are done. Otherwise, at the time when u receives its label, $un(u)$ must be at least 2 since both u and v are unlabeled vertices in $N[u]$, and thus the label of u is at least 2 by line 4 of the algorithm. In this case, v should receive label 0 by line 6 of the algorithm, contradicting that v is an unlabeled vertex. $\square$

Suppose the vertices selected by the algorithm in line 1 and line 3 are $u_1, u_2, \dots, u_p$, in the order of their selection. Notice that all vertices in $\{u_1, \dots, u_p\}$ have positive labels and they form D_R^k . For simplicity of notation, denote by $C[u] = C(u) \cup \{u\}$.

Lemma 2.5 *We have the following properties:*

- (i) $\{C[u_j]\}_{j=1}^p$ is a disjoint partition of V .
- (ii) Node set $\{u_1, \dots, u_p\}$ is an independent set.
- (iii) For any partition of $\{u_1, \dots, u_p\}$ into two disjoint sets U_1 and U_2 , the hop distance between two closest vertices of U_1 and U_2 is two, that is, there is a vertex of label 0 which is adjacent to a vertex in U_1 and a vertex in U_2 .

Proof By Lemma 2.4, $\bigcup_{j=1}^p C[u_j] = V$. For any indices $i \neq j$, we must have $C[u_i] \cap C[u_j] = \emptyset$. This is because every u_j is chosen from currently unlabeled vertices, and it is labeled immediately when it is chosen; only unlabeled vertices are qualified to be put into some $C(u_j)$, and they immediately receive label 0 if the label of u_j is at least 2; in the case that u_j is assigned label 1, u_j is the unique unlabeled vertex in $N[u_1]$ and thus $C(u_j) = \emptyset$. Property (i) is proved.

If property (ii) is not true, consider two adjacent vertices u_i and u_j . Assume, without loss of generality, that $i < j$. Then, when u_i receives its label, vertex u_j is an unlabeled neighbor of u_i . Hence $un(u_i) \geq 2$ at this time point. Since $k \geq 1$, we have $g(u_i) \geq 2$. Then by line 5 to line 7 of the algorithm, vertex u_j will receive label 0, and thus it cannot be chosen in line 3 of the algorithm, a contradiction.

Suppose, without loss of generality, that $u_1 \in U_1$. Let j be the smallest subscript of vertices in U_2 . By line 3 of the algorithm, u_j is adjacent to a vertex v with label 0. Notice that vertex v belongs to $G[\bigcup_{i=1}^{j-1} C[u_i]]$. Hence the parent of v , say u_l , has $l < j$, and thus $u_l \in U_1$. So, v is the vertex satisfying property (iii). $\square$

Lemma 2.6 *The set D_R^k found in line 9 of Algorithm 1 is a strong k -Roman dominating set.*

Proof Suppose $(v_1, \dots, v_k)$ is an arbitrary sequence of vertices arriving online. We are to show that there is always a feasible dispatch scheme. The scheme is as follows: if $v_i = u_j$ for some $j = 1, \dots, p$, then data at v_i can be collected stationary; if $v_i \in C(u_j)$ for some $j = 1, \dots, p$, then a mobile sensor at u_j is moved to collect data at v_i . By Lemma 2.5, such a dispatch scheme can be executed independently for each $C[u_j]$. Notice that the dispatch scheme is feasible if the following property holds all the time:

$$\begin{aligned} &\text{if a job appears at a location of } C(u_j), \\ &\text{node } u_j \text{ has at least two mobile sensors.} \end{aligned} \quad (1)$$

Suppose l_j of the LOIs belong to $C[u_j]$. Notice that $l_j \leq |C[u_j]|$ and $l_j \leq k$. By line 4 of Algorithm 1, u_j is deployed of $\min\{|C[u_j]|, k+1\}$ mobile sensors. If $|C[u_j]| \leq k+1$, then u_j is deployed of $|C[u_j]| \geq l_j$ mobile sensors and thus property (1) holds. If $|C[u_j]| > k+1$, then u_j is deployed of $k+1 > l_j$ mobile sensors. In the case that $k > l_j$, property (1) also holds. Hence, property (1) holds all the time. $\square$

Lemma 2.7 *The set D_{CR}^k output by Algorithm 1 is a connected strong k -Roman dominating set.*

Proof Notice that $D_R^k \subseteq D_{CR}^k$. So, D_{CR}^k is also a strong k -Roman dominating set.

To show the connectedness of $G[D_{CR}^k]$, it suffices to show that in each iteration of the second while loop, as long as $G[D]$ is not connected, there always exists a vertex u with label 0 which is adjacent to R and another connected component of $G[D]$. This is a consequence of property (iii) of Lemma 2.5. $\square$

Theorem 2.8 *For any connected unit ball graph G and any constant integer $k \geq 1$, Algorithm 1 provides a $6(k+2)$ -approximation for the minimum connected strong k -Roman dominating set problem in G .*

Proof Let $D_R^k = (V_1, \dots, V_{k+1})$ be the strong k -Roman dominating set found in line 9 of the algorithm, $D_R^* = (V_{R1}^*, \dots, V_{R,k+1}^*)$ be a minimum strong k -Roman dominating set of G .

Notice that each vertex in $V \setminus V_{R1}^*$ either belongs to $V_{R2}^* \cup \dots \cup V_{R,k+1}^*$ (which has label at least 2) or is adjacent to some vertex of $V_{R2}^* \cup \dots \cup V_{R,k+1}^*$ (which has label 0). In other words, each vertex in $V \setminus V_{R1}^*$ is dominated by some vertex of $V_{R2}^* \cup \dots \cup V_{R,k+1}^*$. By Lemma 2.5 (ii), D_R^k is an independent set. By Proposition 2.3, every vertex in $V_{R2}^* \cup \dots \cup V_{R,k+1}^*$ can dominate at most 12 independent vertices.

Hence

$$|D_R^k| - |V_{R1}^*| \leq |D_R^k \setminus V_{R1}^*| \leq 12(|V_{R2}^*| + \dots + |V_{R,k+1}^*|).$$

It follows that

$$\begin{aligned} w(D_R^*) &= |V_{R1}^*| + 2|V_{R2}^*| + 3|V_{R3}^*| + \dots + (k+1)|V_{R,k+1}^*| \\ &\geq |V_{R1}^*| + 2(|V_{R2}^*| + |V_{R3}^*| + \dots + |V_{R,k+1}^*|) \\ &\geq |V_{R1}^*| + 2 \times \frac{1}{12} (|D_R^k| - |V_{R1}^*|) \\ &= \frac{1}{6}|D_R^k| + \frac{5}{6}|V_{R1}^*| \geq \frac{1}{6}|D_R^k|, \end{aligned}$$

By Lemma 2.5 (iii), to change D_R^k into D_{CR}^k , it is sufficient to change the labels of at most $|D_R^k| - 1$ vertices from 0 to 1. Hence

$$\begin{aligned} w(D_{CR}^k) &\leq w(D_R^k) + |D_R^k| - 1 \\ &\leq (|V_1| + 2|V_2| + \dots + (k+1)|V_{k+1}|) + (|V_1| + |V_2| + \dots + |V_{k+1}|) \\ &\leq (k+2)|D_R^k| \\ &\leq 6(k+2)w(D_R^*) \\ &\leq 6(k+2)w(D_{CR}^*), \end{aligned}$$

where D_{CR}^* is a minimum connected strong k -Roman dominating set, whose weight is clearly an upper bound for $w(D_R^*)$. The theorem is proved. $\square$

Theorem 2.9 *The time complexity of Algorithm 1 is $O(mn)$, where m and n are the number of edges and the number of vertices in the unit ball graph, respectively.*

Proof In each iteration of the first while loop, at least one vertices gets its label. Hence the first while loop is executed at most n rounds. And in each round, a breadth first search might be used, which takes time $O(m)$. Similarly, the second while loop also takes time $O(mn)$.

3 Conclusion

In this paper we propose the connected strong k -Roman dominating set for the purpose of modeling the defence problem of Roman Empire in the presence multiple attacks, and present a constant approximation algorithm with a guaranteed performance ratio at most $6(k + 2)$ in unit ball graphs. It is interesting and remains open how to obtain a better approximation in wireless sensor networks for future research.

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